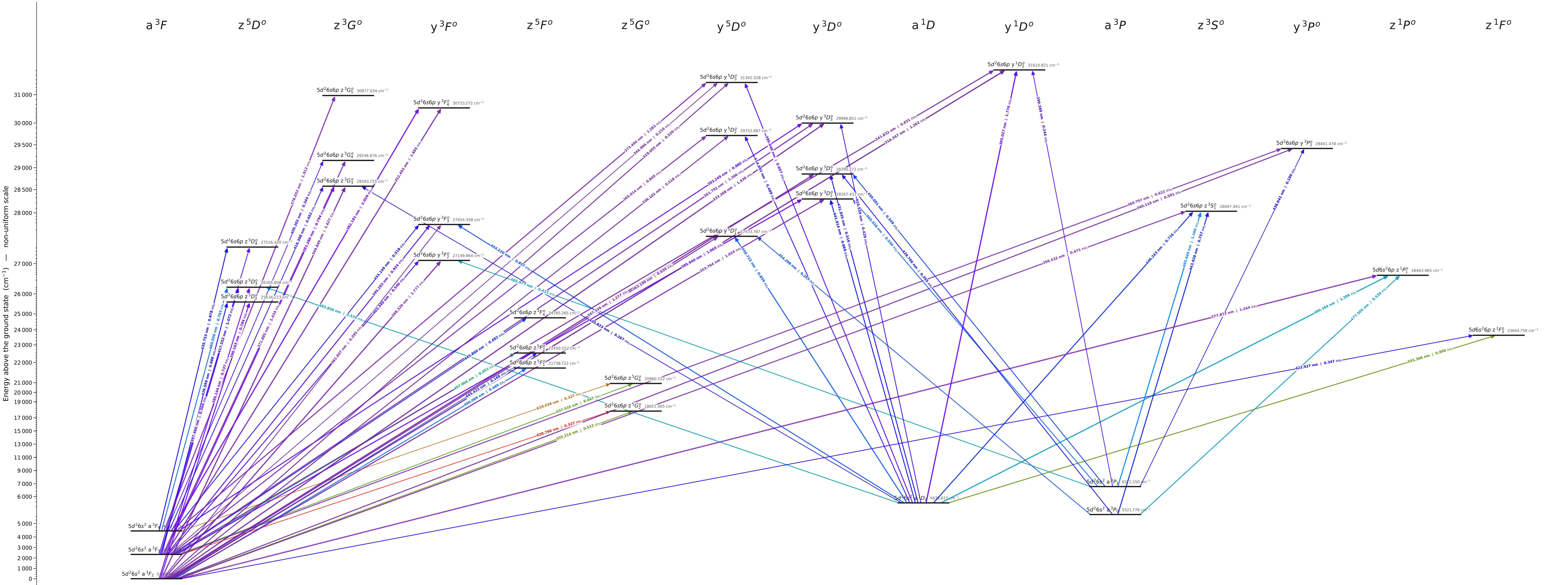
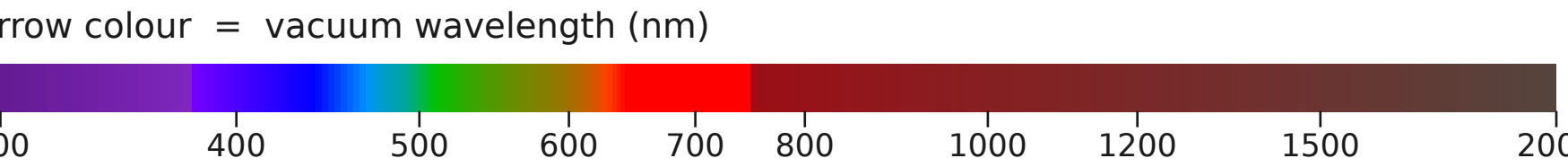


Hafnium

Hf (Hf I). Strongest classified transitions below 2 μm (NIST): 31 levels, 70 lines.

Hf⁺ ionisation limit 55047.90 cm⁻¹ = 6.82507 eV (181.660 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle [J||e\mathbf{r}||J'] \rangle$ in $e a_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, = = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle [J e\mathbf{r} J'] \rangle$ (eao)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
5d ² 6s ² a ³ F ₂ – 5d ² 6s6p y ¹ D ₂ ^o	316.3474	316.2558	947.6686	1.262	2.040e+07	–	–	–
5d ² 6s ² a ³ F ₂ – 5d ² 6s6p y ³ D ₂ ^o	333.3683	333.2725	899.2830	1.530	1.830e+07	–	–	–
5d ² 6s ² a ³ F ₂ – 5d ² 6s6p y ³ D ₁ ^o	347.3395	347.2401	863.1106	1.277	2.630e+07	–	–	–
5d ² 6s ² a ³ F ₂ – 5d ² 6s6p z ³ G ₂ ^o	349.8494	349.7494	856.9185	1.627	1.790e+07	–	–	–
5d ² 6s ² a ³ F ₃ – 5d ² 6s6p y ³ F ₄ ^o	352.4031	352.3023	850.7090	1.685	1.460e+07	–	–	–
5d ² 6s ² a ³ F ₃ – 5d ² 6s6p y ³ D ₂ ^o	353.7639	353.6629	847.4364	1.024	9.600e+06	–	–	–
5d ² 6s ² a ³ F ₃ – 5d ² 6s6p y ³ D ₁ ^o	361.7922	361.6891	828.6316	1.366	1.140e+07	–	–	–
5d ² 6s ² a ³ F ₂ – 5d ² 6s6p y ³ F ₂ ^o	368.3287	368.2238	813.9265	1.777	2.560e+07	–	–	–
5d ² 6s ² a ³ F ₃ – 5d ² 6s6p z ³ G ₂ ^o	371.8855	371.7797	806.1419	1.434	9.000e+06	–	–	–
5d ² 6s ² a ³ F ₄ – 5d ² 6s6p y ³ D ₂ ^o	373.4843	373.3781	802.6909	1.283	9.140e+06	–	–	–
5d ² 6s ² a ³ F ₂ – 5d6s ² 6p z ¹ P ₁ ^o	377.8723	377.7650	793.3697	1.264	2.000e+07	–	–	–
5d ² 6s ² a ³ F ₄ – 5d ² 6s6p z ³ G ₂ ^o	378.6532	378.5456	791.7337	1.912	1.240e+07	–	–	–
5d ² 6s ² a ³ F ₄ – 5d ² 6s6p y ³ F ₂ ^o	382.1807	382.0723	784.4259	2.059	1.710e+07	–	–	–
5d ² 6s ² a ¹ D ₂ – 5d ² 6s6p y ¹ D ₂ ^o	385.0270	384.9178	778.6272	1.776	2.240e+07	–	–	–
5d ² 6s ² a ³ P ₁ – 5d ² 6s6p z ³ S ₁ ^o	465.6493	465.5189	643.8160	1.160	9.000e+06	–	–	–
5d ² 6s ² a ¹ D ₂ – 5d6s ² 6p z ¹ P ₁ ^o	480.1840	480.0498	624.3283	1.299	1.030e+07	–	–	–

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).